

Grievance Expression and Collective Discourse in Instagram Protest Communication: Evidence from the 2025 Indonesian Protests

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Abstract. Grievances play a central role in collective action, yet their dynamics in online protest environments remain difficult to measure at scale. This study presents a socio-computational framework for analyzing grievance discourse during the 2025 Indonesian protests on Instagram. Using 18,825 Instagram posts, we combine large language model (LLM)-based grievance detection, collective identity analysis, linguistic signal analysis, and topic-based network analysis. Evaluation against manually annotated data demonstrates reliable grievance identification, achieving 80% accuracy and a Cohen’s κ of 0.589. Results show that grievance discourse is associated with stronger collective identity framing and greater emotional expression than non-grievance discourse. Political and social grievances further exhibit larger and more cohesive topic-centered discussion communities than economic grievances, particularly during periods of heightened protest activity. Together, these findings demonstrate the value of computational approaches for understanding grievance expression and collective discourse in online social movements.

Keywords: Grievance Analysis · Social Movements · Protest Dynamics · Large Language Models · Instagram

1 Introduction

Social media platforms have become important spaces for expressing grievances, constructing collective identities, and facilitating protest-related communication. During periods of political unrest, online discourse provides insight into how dissatisfaction is framed, shared, and interpreted within collective action [13, 14].

The 2025 Indonesian protests provide an important case for examining these dynamics. What began as criticism of parliamentary housing allowances expanded into broader concerns regarding corruption, governance, inequality, and police violence. Public mobilization intensified following the death of Affan Kurniawan during a police intervention on August 28, transforming localized protests into a nationwide movement. Although the government revoked the housing allowance policy, demonstrations continued under a broader reform agenda, re-

flecting sustained public grievances. Instagram served as a major platform for expressing these grievances and supporting protest-related discourse.

In this study, we present a socio-computational framework for analyzing grievance discourse on Instagram. Using large language models (LLMs), collective identity analysis, linguistic signal analysis, and topic-based network analysis, we examine how grievances evolve, how they relate to collective identity and linguistic characteristics, and how different grievance domains are reflected in topic-based discussion structures. Specifically, we address the following research questions: **RQ1:** How effectively can large language models identify grievance-related discourse at scale, and how do grievance dynamics evolve throughout the protest period? **RQ2:** How do grievance and non-grievance posts differ in their collective identity framing and linguistic characteristics? **RQ3:** How do political, social, and economic grievance domains differ in their topic-based discussion structures?

2 Literature Review

Grievance is widely recognized as a key driver of collective action, although dissatisfaction alone does not automatically lead to protest participation. Mobilization depends not only on perceived injustice but also on the availability of opportunities, resources, and effective channels for action [7, 2]. Research further suggests that grievances are socially constructed through collective interpretation of political and social events rather than objective conditions alone [4, 12].

Collective identity provides an important link between grievances and collective action by transforming individual concerns into shared group experiences that encourage participation [8]. Similar processes have been observed in online environments, where social media enables large-scale participation while protest discourse remains fragmented across communities and interpretations [11].

Computational approaches have examined protest communication using topic modeling, sentiment analysis, and interaction networks [6]. More recent NLP studies frame grievance as a text classification task using large language models [10], with greater emphasis on model performance than on grievance dynamics and collective discourse.

Motivated by these gaps, we extend our previous grievance detection framework [1] to Instagram, examining the relationship between grievance, collective identity, linguistic behavior, and topic-based discourse during the 2025 Indonesian protests.

3 Methodology

We integrate LLM-based grievance detection, collective identity analysis, linguistic characterization, and network analysis to examine grievance expression and collective discourse.

3.1 Data Collection

Instagram posts related to the 2025 Indonesian protests were collected using the APIFY platform through a keyword-based search. The initial query was constructed from news reports and protest coverage, including prominent hashtags such as #IndonesiaGelap, #TolakRUUTNI, #SupremasiSipil, #BubarkanDPR, and #RakyatTagihJanji. The query was subsequently expanded using a snowballing approach based on frequently co-occurring hashtags and protest-related terms.

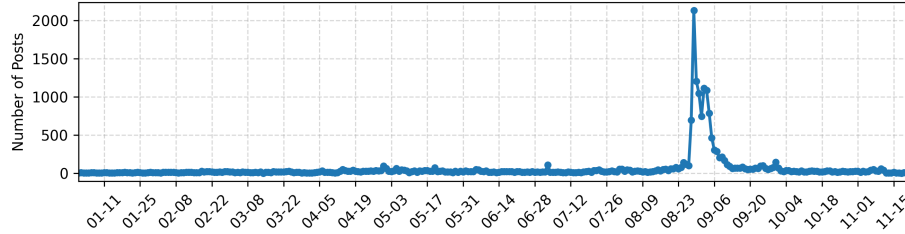


Fig. 1: Temporal distribution of Instagram posts related to the 2025 Indonesian protests. Daily post counts reveal a pronounced increase in activity during the late-August protest escalation.

The final dataset spans January 1 to November 21, 2025 and contains 18,825 Instagram posts with captions, hashtags, timestamps, and engagement meta-data. Figure 1 shows the temporal distribution of posts, with activity peaking during the late-August protest escalation. Because many captions were written in Indonesian, all posts were translated into English using `gpt-oss-120b`, selected for its multilingual reasoning capability and consistent structured outputs. Translation quality was evaluated through back-translation on a random sample of 300 posts. Using `qwen3-embedding-8b` embeddings, the translated texts achieved an average cosine similarity of 0.89, indicating strong semantic preservation.

3.2 LLM-Based Grievance Detection

To address **RQ1**, we formulate grievance identification as a binary classification task in which each Instagram post is labeled as either *grievance* or *non-grievance*. Following our previously proposed grievance detection framework [1], grievance is defined as expressions of perceived injustice, harm, unfairness, corruption, repression, or demands for accountability. We use the open-weight `gpt-oss-120b` model to classify translated Instagram captions through structured prompting. The prompt explicitly distinguishes grievance from related discourse, including emotional reactions, neutral reporting, humor, and mobilization without explicit attribution of wrongdoing, and returns a binary label for each post.

To evaluate reliability, we conducted a human–LLM agreement study on a random sample of 300 Instagram posts. Three annotators independently labeled each post following the same annotation guidelines, with majority voting used to construct the gold-standard labels. LLM predictions were evaluated using accuracy, precision, recall, F1-score, and Cohen’s κ .

3.3 Collective Identity Analysis

To address **RQ2**, we examine collective identity formation through self-referential pronouns in the translated captions, an established proxy in computational studies of social movements [8]. Singular identity terms include first-person singular references (e.g., “I”, “me”, “my”), whereas collective identity terms include first-person plural references (e.g., “we”, “us”, “our”). Collective identity is computed as the proportion of collective references relative to the total number of collective and singular references: $CollectiveIdentity = \frac{Collective}{Collective + Singular}$. Temporal trends are subsequently compared between grievance and non-grievance posts.

3.4 Linguistic Signals as Proxy for Mobilization

To further address **RQ2**, we compare grievance and non-grievance posts using four linguistic signal categories proposed by Amure and Agarwal [3]: conative, referential, emotive, and expressive. Conative signals represent action-oriented language, referential signals capture informational content, emotive signals reflect affective reactions, and expressive signals capture subjective evaluations. Linguistic signals are automatically identified using the computational framework introduced by [3].

For each caption, normalized signal ratios are computed relative to the total number of words, $Ratio = \frac{ExpressionCount}{TotalWords}$. Word counts are obtained using the spaCy `en_core_web_trf` model.

3.5 Grievance Domains and Co-Topic Network Construction

To address **RQ3**, grievance posts are assigned to one or more domains (political, social, and economic) using LLM-generated confidence scores. A confidence threshold of 0.7 is adopted to retain high-confidence assignments while allowing posts to belong to multiple grievance domains when appropriate.

For each domain, BERTopic [5] clusters semantically similar posts using transformer embeddings, after which related topics are merged into 20 broader themes to improve interpretability. Weekly co-topic networks are then constructed by connecting users who discuss at least one common topic during the same week. Consequently, network edges represent thematic similarity rather than direct user interaction.

To quantify discourse organization, modularity is computed using NetworkX. Higher modularity indicates stronger separation between topic-centered discussion communities.

4 Results

Results are organized around the three research questions. We first evaluate grievance detection, followed by analyses of collective identity, linguistic signals, and grievance-specific discourse networks.

4.1 Grievance Detection and Patterns in Movement

Table 1 shows that the proposed approach achieves 80% accuracy, an F1-score of 0.84 for grievance posts, and a mean Cohen’s κ of 0.589, indicating moderate agreement between human annotators and the LLM. The high precision for grievance posts suggests that the model effectively identifies discourse centered on perceived injustice, harm, and accountability. These results support the use of LLM-based grievance detection for large-scale analysis of protest discourse, thereby addressing **RQ1**.

Table 1: Classification report with Cohen’s κ of the human-LLM agreement for binary grievance detection. Labels were constructed using majority voting across three human annotators.

Class	Precision	Recall	F1-Score	Support
No Grievance	0.62	0.90	0.74	92
Grievance	0.95	0.76	0.84	208
Accuracy			0.80	300
Macro avg	0.79	0.83	0.79	300
Weighted avg	0.85	0.80	0.81	300
Mean Cohen’s Kappa: 0.589				

Figure 2 shows that grievance expression remained a persistent feature of protest discourse, with an average grievance rate of 63% and more than 11,000 posts receiving grievance scores above 70%. Throughout the observation period, discussions were consistently framed through narratives of injustice, accountability, and dissatisfaction. Grievance levels further increased during the late-August protest escalation following the death of Affan Kurniawan, indicating that major offline events intensified existing grievance narratives rather than fundamentally changing the nature of the discourse.

4.2 Collective Identity and Grievance Discourse

Figure 3 shows that grievance discourse consistently exhibits stronger collective identity framing than non-grievance discourse, suggesting that grievances are more frequently expressed as shared group experiences rather than individual concerns. This finding supports **RQ2** and is consistent with the role of collective identity in mobilizing individuals around common perceptions of injustice. To illustrate this pattern, a representative grievance post from a period of heightened

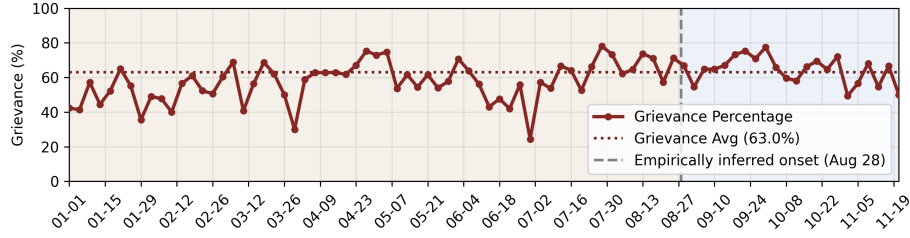


Fig. 2: Daily percentage of posts classified as grievances. The dashed line denotes the average grievance rate (63%) across the dataset.

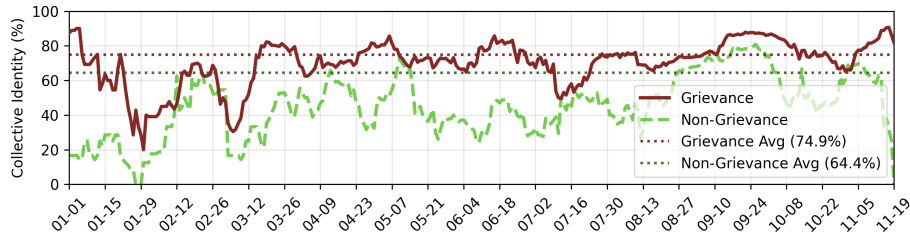


Fig. 3: Collective identity over time, measured as the proportion of collective references relative to collective and singular self-references. Higher values indicate stronger collective identity framing.

collective identity framing (August 28–September 1) is shown below. Collective references are highlighted in **blue**, while singular self-references are highlighted in **orange**.

- **Grievance:** To all the tie-wearing rats who dance on the misery of others, **I** hope karma finally catches up with you. For all the lives you took, for all the voices you silenced, for all the passions you tried to extinguish, **I** hope you realize that **we** are all human beings. **We** may be just ordinary people who have no power like you, but **we** will not retreat. **We** will move forward together. **we** will not give up. May you be free, **my** Indonesia. #bubarkandpr

4.3 Linguistic Signals as Mobilization Indicators

We next address the linguistic component of **RQ2** by comparing linguistic signals associated with grievance and non-grievance posts. Figure 4 shows that grievance posts consistently contain lower levels of conative and referential language but higher levels of emotive and expressive language than non-grievance posts. This pattern suggests that grievance discourse primarily communicates dissatisfaction and perceived injustice, whereas non-grievance discourse emphasizes information sharing and calls for action, indicating that grievance expression and mobilization perform complementary roles within online protest discourse.

The divergence becomes particularly evident during the late-August protest escalation following the death of student protester Affan Kurniawan in Jakarta.

During this period, emotive and expressive language increased with heightened grievance activity, reflecting public reactions to police violence, accountability, and perceived injustice. In contrast, non-grievance posts exhibited higher conative and referential signals, indicating efforts to disseminate information, coordinate demonstrations, and encourage participation. Together, these findings suggest that major offline protest events influenced both the volume of online discussion and the linguistic strategies through which participants expressed grievances and supported collective action [15, 9].

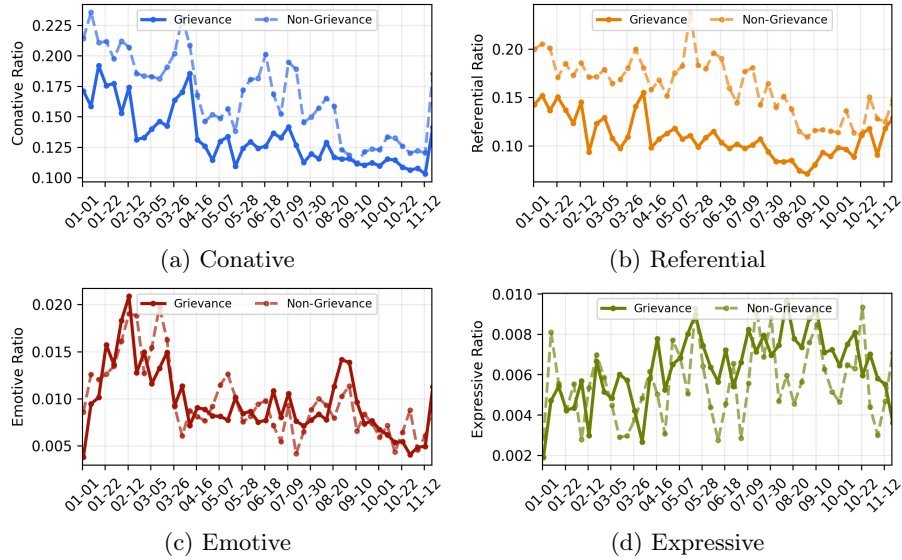


Fig. 4: Temporal trends in conative, referential, emotive, and expressive language. Values represent average post-level linguistic ratios computed over time.

To illustrate these patterns, a representative grievance post from the late-August protest escalation (August 28–September 1) is shown below. Conative and referential terms are highlighted in blue and orange, respectively, while emotive (EMO) and expressive (EXP) signals are underlined and annotated at the sentence level. Since emotive and expressive signals are extracted at the sentence level, their normalized ratios are not directly comparable with the word-level conative and referential ratios.

- **Grievance:** It feels like I can no longer trust this country.^{EXP}
Tired of being an Indonesian citizen.^{EXP} Tired of witnessing
the stupidity and barbarity displayed by the power holders every day.^{EMO} A
collective wound that grows every second, who knows when it will heal.
Angry Sad.^{EMO} Heartbroken.^{EMO} But the resistance must not stop. Long
live the resistance. The martyrs who passed away will never be forgotten.^{EMO}

P.S. I took the photos from Threads, I am not the owner. #menolaklupa
#indonesiagelap #ripindonesia

4.4 Grievance Domains and Network Structure

To address **RQ3**, we analyze grievance categories across time and network structure. Figure 5 shows that political and social grievances dominate protest discussions, while economic grievances remain comparatively limited. The late-August surge across all categories indicates that protest discourse expanded beyond the initial economic concerns to broader issues of governance, accountability, and social justice.

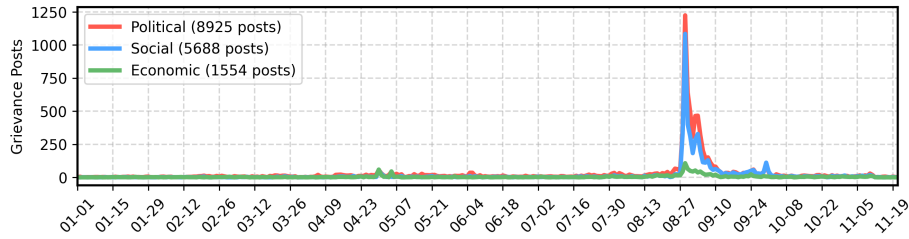


Fig. 5: Daily volume of posts assigned to political, social, and economic grievance categories. A post may belong to multiple categories.

Table 2: Descriptive statistics of grievance-domain co-topic networks.

Network	Nodes	Edges	Density
Political	2,996	1,429,998	0.32
Social	2,445	746,656	0.25
Economic	659	23,987	0.11

The network statistics in Table 2 show that political and social grievances generate substantially larger and denser discussion communities than economic grievances. Figure 6 further reveals relatively high modularity across grievance domains during major protest periods, indicating the emergence of distinct topic-centered communities rather than a single homogeneous discussion space. Together, these findings suggest that grievance domains differ not only in discussion volume but also in how they organize collective discourse, with political and social grievances supporting larger and more cohesive topic-centered communities than economic grievances.

To illustrate the overlap between grievance domains, a representative post assigned to political, economic, and social grievance categories is presented. The post simultaneously criticizes government actions, economic inequality, and the

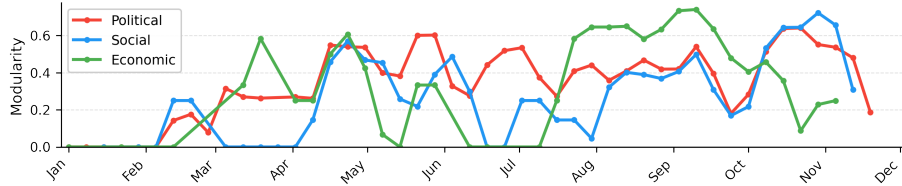


Fig. 6: Weekly modularity of co-topic user networks across three grievance domains. Higher values indicate stronger community structure.

perceived neglect of citizens, demonstrating how multiple grievance types can co-occur within protest discourse:

- “While millions of people are still struggling to find food, enduring hunger, and finding it difficult to get jobs, members of the DPR instead receive a salary of Rp 3 million per day. How cruel is the irony of this country: the people are forced to survive on scraps of rupiah, while the people’s representatives party with the people’s money. What kind of justice is this, when the servants of the nation live like kings, while the nation they serve is left to suffer?” #bubarkandpr #fightordie

5 Conclusion and Future Work

This study presented a socio-computational framework for analyzing grievance expression and collective discourse during the 2025 Indonesian protests on Instagram. Using LLM-based grievance detection, collective identity analysis, linguistic characterization, and topic-based network modeling, we showed that grievance discourse is strongly associated with collective identity framing and emotional expression. In contrast, non-grievance discourse exhibits stronger informational and action-oriented language, suggesting that grievance expression and mobilization perform complementary roles within protest communication. We further demonstrated that political and social grievances generate larger and more cohesive thematic communities than economic grievances, highlighting their role in structuring collective discourse during political contention. More broadly, the findings demonstrate how computational approaches can identify not only when grievances emerge but also how they are collectively framed and organized during protest movements.

This study has several limitations. Grievance domain assignment relies on LLM confidence scores, collective identity is approximated through lexical self-references, and the co-topic network captures thematic similarity rather than direct user interactions. Furthermore, our analysis focuses on Instagram and a single protest movement, which may limit the generalizability of the findings. Despite these limitations, the proposed framework provides a scalable approach for studying online protest discourse. Future work will extend the framework to multimodal content, cross-platform protest ecosystems, and comparative analyses across social movements.

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